Journal of Electromyography and Kinesiology 23 (2013) 1485–1493

Contents lists available at ScienceDirect

Journal of Electromyography and Kinesiology

journal homepage: www.elsevier.com/locate/jelekin

Muscle activation and coactivation during five-time-sit-to-stand movement in patients undergoing total knee arthroplasty

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ARTICLE INFO

Article history:
Received 20 November 2012
Received in revised form 24 April 2013
Accepted 7 June 2013

Keywords:
TKA
Coactivation
Sit to stand
Quadriceps strength
Osteoarthritis
Electromyography

ABSTRACT

Quadriceps weakness is prevalent with knee osteoarthritis (OA) and after total knee arthroplasty (TKA). To compensate for quadriceps dysfunction, patients often alter movement strategies. Little is known about muscle coordination during sit-to-stand (concentric) and stand-to-sit (eccentric) movements in the acute postoperative period. This investigation characterized the distribution of muscle activation between the concentric and eccentric phases during a five-time-sit-to-stand (FTSTS) movement in late stage OA and one month after TKA. Patients and healthy participants performed a FTSTS while recording bilateral ground reaction forces (GRFs) and electromyography (EMG). Concentric and eccentric ensemble averages of the GRF and EMG were calculated for the concentric and eccentric phases. Coactivation indices, integrated EMG, and GRF were calculated for each limb and phase. Patients demonstrated higher eccentric coactivation than the healthy group. Postoperative loading was higher in the nonsurgical limb. Postoperative quadriceps activity was lower in the concentric phase and higher in the eccentric phase than the healthy group. Higher coactivation in the patients resulted from sustained distribution of quadriceps activity throughout the eccentric phase. This indicated an inability to coordinate muscle firing when rapidly lowering to a chair and occurred despite unloading of the surgical limb. Although these patterns may serve as a protective strategy, they may also impede recovery of muscle function after TKA.
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1. Introduction

Symptomatic knee osteoarthritis (OA) affects over 12% of adults ages 60 and over in the United States (Dillon et al., 2006). Total knee arthroplasty (TKA) is frequently performed to relieve pain and disability in patients suffering from end-stage knee OA. Although TKA reliably reduces pain, a combination of strength and functional performance deficits may persist several years after surgery (Ritter et al., 2004; Silva et al., 2003).

Compared with their healthy counterparts, patients with knee OA demonstrate lower extremity muscle weakness that is exacerbated one month after surgery (Bade et al., 2010). In particular, quadriceps strength deficits of 50–60% exist one month postoperatively (Mizner et al., 2005; Stevens-Lapsley et al., 2010) and strength deficits persist years after surgery (Silva et al., 2003). Quadriceps weakness and central activation deficits are associated with declines in functional performance in patients following neurologic injury (Horstman et al., 2008) as well as frail, older adults (Brown et al., 1995). To accommodate for this muscle weakness, patients frequently alter muscle activation and movement strategies during functional tasks both before and after TKA. For example, patients often employ a "knee stiffening strategy" during walking in which they limit their sagittal plane knee range of motion and concurrently reduce the knee extension moment while increasing quadriceps/hamstrings muscle coactivation (Benedetti et al., 2003).

Movement strategies by which patients limit quadriceps work have also been measured in other activities of daily living. For example, Mizner and Snyder-Mackler (2005a) linked quadriceps weakness to difficulties in performing a sit-to-stand task. Three months after TKA, patients performed a sit-to-stand task with symmetrical sagittal plane joint kinematics, but with smaller peak vertical ground reaction forces and lower quadriceps muscle recruitment in the surgical limb compared to the nonsurgical limb. These asymmetric knee joint kinetics were associated with asymmetric quadriceps strength between limbs (Mizner and Snyder-Mackler, 2005b). Further, when compared to healthy

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1050-6411/$ - see front matter © 2013 Published by Elsevier Ltd.
http://dx.doi.org/10.1016/j.jelekin.2013.06.008

adults, patients after TKA performed the sit-to-stand movement with larger hip flexion angles (Farquhar et al., 2008), which reduces the required knee extension moment during a sit to stand activity by positioning the center of mass of the trunk closer to the knee joint center (Doorenbosch et al., 1994). Similar to a stiff knee gait, this strategy reduces the mechanical work performed by the quadriceps, perhaps as a response to the existing quadriceps muscle weakness. Although this strategy successfully reduces the demand on the quadriceps, it also increases the work performed by the hip extensor muscles (Farquhar et al., 2008), which includes the hamstrings and may contribute to altered muscle coactivation strategies about the knee joint. The combination of increasing hip flexion and muscle coactivation about the knee results in difficulty isolating the quadriceps muscle during what is normally a quadriceps-dominated task. This lack of isolated force production by the quadriceps during daily activity may prove detrimental to recovery of normal quadriceps muscle function (Rossi et al., 2006; Silva et al., 2003; Walsh et al., 1998).

Although altered movement strategies during sit-to-stand tasks after TKA have been examined in previous work, less is known about the muscle activation patterns during this task, especially in the acute postoperative period. In addition, muscle activation patterns while lowering to a chair before and after TKA are unknown. Like the sit-to-stand task, lowering to a chair demands larger joint moments at the hip and knee than other activities of daily living (Doorenbosch et al., 1994). However, lowering the body requires a controlled deceleration of the center of mass against gravity. This deceleration is accomplished through eccentric activation of the lower extremity extensors, which may be impaired in the presence of quadriceps dysfunction. Because lowering to a chair places different demands on the knee joint and the surrounding musculature than the sit-to-stand task, examining muscle activity patterns during each task may provide unique insights into coordination, coactivation, and compensatory strategies in patients before and after TKA.

The primary objective of this study was to characterize muscle activations during a sit-to-stand task and during lowering to a chair in patients with unilateral TKA and healthy adults of similar age. We examined bilateral muscle activity in the quadriceps and hamstrings using electromyography (EMG), the ground reaction force under each limb during the task, the time to complete the five-time-sit-to-stand task, and muscle strength. We hypothesized that patients would demonstrate asymmetric limb loading and slower performance of a five-time-sit-to-stand task before and after TKA compared with healthy adults. We also hypothesized that patients would demonstrate greater levels of surgical limb quadriceps/hamstring coactivation during the lowering (eccentric) phase of the task both before and after TKA when compared to the healthy group. Finally, we hypothesized that patients would perform a five-time-sit-to-stand task more slowly, with more asymmetry, and with higher levels of quadriceps/hamstrings coactivation one month after surgery compared with before surgery.

2. Methods

2.1. Participants

This investigation included two groups: (1) patients with end-stage OA scheduled to receive a unilateral TKA and (2) healthy participants (Table 1), and consisted of a sample of convenience. Patients with end-stage OA ($n = 10$) performed maximum isometric knee extensor and flexor strength testing and a five-time-sit-to-stand (FTSTS) task before receiving a primary, unilateral quadriceps splitting TKA (preoperative). Patients were excluded from the study if pain in the contralateral knee was greater than 4/10 during activity or if any other unstable lower-extremity orthopedic conditions were present. One month after TKA (postoperative), patients visited the laboratory again for the same testing. Healthy participants ($n = 13$) performed the FTSTS task on one occasion in an identical session as the patient group. This protocol was approved by the Colorado Multiple Institutional Review Board (COMIRB), and each participant provided written informed consent.

Table 1
Mean (SD) of baseline anthropometry. Age, height, mass, and BMI were tested with two-tailed paired *t*-tests and distribution of sex was tested with a two-tailed Chi-squared test.

	Patient	Healthy	*P*-value
Age (years)	59.7 (9.1)	63.9 (6.8)	0.22
Height (m)	1.70 (0.12)	1.73 (0.09)	0.50
Mass (kg)	84.4 (17.0)	79.8 (15.0)	0.28
BMI (kg/m^2)	28.0 (4.1)	26.4 (3.8)	0.39
% Female	70.0	38.4	0.13
% Male	30.0	61.6	

2.2. Strength testing

Maximum voluntary isometric (MVIC) quadriceps and hamstrings muscle strength measurements were performed while participants were seated and stabilized on an electromechanical dynamometer (HUMAC NORM CSMI, Stoughton, MA). Participants were positioned in 85° of hip flexion and 60° of knee flexion as previously described (Mintken et al., 2007; Stevens-Lapsley et al., 2010). Data were sampled using a Biopac Data Acquisition System at a sampling frequency of 2000 Hz (Biodex Medical Systems, Inc., Shirley, NY) and analyzed using AcqKnowledge software, Version 3.8.2 (Biodex Medical Systems). Strength measurements were expressed in units of normalized torque (N m/kg), which was calculated by multiplying the force (N) collected by the dynamometer by the lever arm length (m; medial joint line to one inch superior to the medial malleolus) and normalizing by mass (kg). All participants were given visual targets and strong verbal encouragement during each MVIC to achieve an accurate maximum effort. MVICs for each muscle group were performed twice; however, if maximum torque during the first two trials differed by more than 5%, a third trial was performed. The trial with the highest torque was used for analysis.

2.3. Five-time-sit-to-stand task

During the FTSTS task, participants were seated in a chair of standardized seat height (46 cm) with their feet placed on the center of each force platform (Whitney et al., 2005). They were instructed to perform the FTSTS task "as quickly and safely as possible". Time taken to transfer five times between the sitting and standing positions was recorded. Participants were encouraged to avoid using their hands during the test, both on the chair armrests and their legs. However, nine patients used the armrests for assistance during preoperative collection and eight patients used the armrests for assistance during postoperative collection. To preserve natural movement, no additional instructions were given.

Ground reaction forces (GRFs) and electromyography (EMG) data were recorded during the FTSTS task. Vertical GRFs under each limb were recorded using individual force platforms (PASCO scientific, Roseville, CA) placed under each foot. After cleaning the skin surface with isopropyl alcohol, differential bilateral surface EMG electrodes were placed on quadriceps (*vastus lateralis*) and hamstrings (*biceps femoris*) according to SENIAM guidelines (Merletti and Hermens, 2000). These active electrodes were Delsys

DE2.3 electrodes with Ag bars (inter-electrode distance: 1 cm, CMRR > 100 dB). Signals were amplified by 1000 and stored using the Myomonitor IV datalogger (Delsys Inc., Boston, MA). Data from the two sources were synchronized by aligning temporal features of Myomonitor pressure sensors (placed under each heel) with the GRF.

2.4. Data processing

The FTSTS task was partitioned into five repetitions. Because kinematic data were not available, each repetition was separated into a sit-to-stand (concentric) and a stand-to-sit (eccentric) phase using the GRF as follows: GRF from the surgical and nonsurgical limbs were added to provide a combined GRF during the FTSTS task. Acceleration of the center of mass is proportional to the change in vertical GRF (Winter, 2009). This relation indicates that the eccentric and concentric phases can be divided at the local maximum of the vertical GRF (Fig. 1). This method of dividing each repetition into concentric and eccentric phases was validated by knee angle and GRF data collected from two participants in a fully instrumented human movement laboratory.

Outcome measures from the FTSTS task were calculated for each phase (concentric, eccentric) of the GRF and EMG data. GRF amplitudes from each repetition were smoothed using a zero phase-lag Butterworth filter with 10 Hz cutoff frequency, which represents approximately 98% of the power in the frequency spectrum, and then normalized by the peak value of the repetition. Ensemble averages were calculated for the concentric and eccentric phases. To characterize loading of each limb, the ensemble averages were time normalized to repetition and integrated:

$$\mathrm{iGRF} = \int_{t_0=0\%}^{t_f=100\%} \mathrm{GRF}\,dt \tag{1}$$

where *GRF* is the normalized ground reaction force under the nonsurgical limb, surgical limb, or sum of the two limbs. This integrated GRF (iGRF) represents the impulse under each limb, and represents the contribution of the limb to the change in vertical momentum (acceleration/deceleration) of the center of mass (COM) (Robertson et al., 2004). The time bounds used for all integrals are the beginning (0%) and the end (100%) of each phase (concentric, eccentric).

EMG data were bandpass filtered using a 4th order zero phase-lag Butterworth filter with 20–350 Hz cutoff frequencies (Merletti, 1999). The Teager-Kaiser Energy Operator (TKEO), a transformation that combines the amplitude and instantaneous frequency (Hug, 2011), was applied to the bandpass filtered signal. Representing muscle activation with TKEO when low signal-to-noise ratios are encountered is advisable in onset detection (Li and Aruin, 2005) and was used to represent dynamic EMG signals (Prosser et al., 2010). Signals were smoothed to create a linear envelope using full-wave rectification and applying a 4th order zero phase-lag Butterworth filter (6 Hz low-pass cutoff) which is similar to typical EMG processing for gait signals (Burden, 2010; Yang and Winter, 1984).

To reduce inter-individual variability of the data, linear envelop EMG data from each repetition were normalized by dividing by the mean EMG amplitude of the repetition (Burden, 2010; Yang and Winter, 1984). Normalization was applied to each repetition separately to eliminate potentially confounding EMG signals from other repetitions in the FTSTS task. An average of the five repetitions was calculated (Fig. 2) and separated into concentric and eccentric phases before calculating dependent variables.

Muscle activity was quantified for each phase (concentric, eccentric) by integrating the averaged EMG data:

$$\mathrm{iEMG}_{\mathrm{muscle}} = \int_{t_0=0\%}^{t_f=100\%} \mathrm{nEMG}_{\mathrm{muscle}}\,dt \tag{2}$$

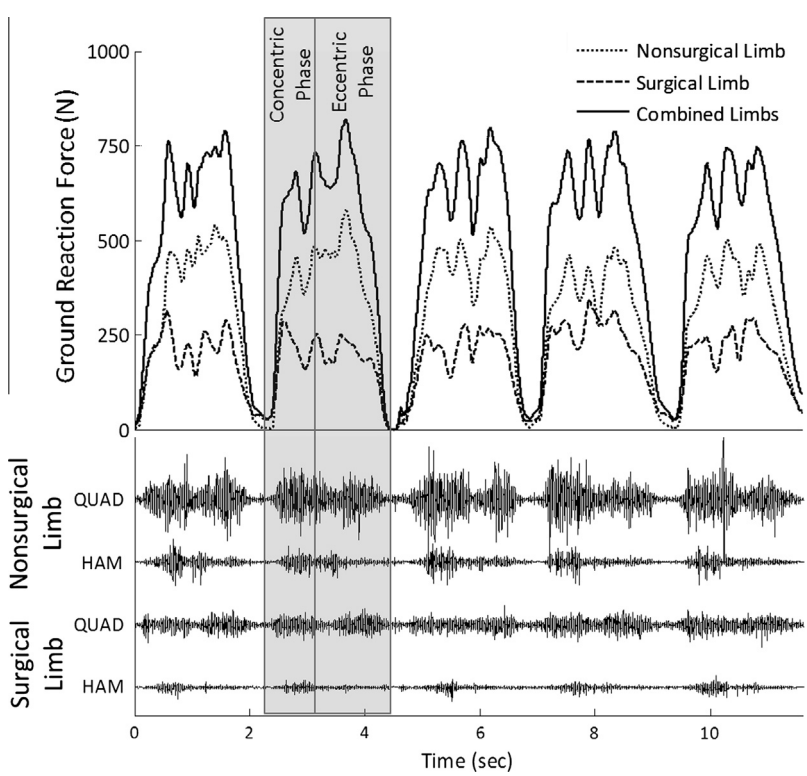

Fig. 1. Example postoperative ground reaction forces and raw EMG signals from the nonsurgical and surgical limb during the FTSTS task. Vertical ground reaction forces from each limb were combined, and the submaximal peak in the middle of the repetition, which indicates the beginning of downward velocity, was used to separate the concentric and eccentric phases of each repetition.

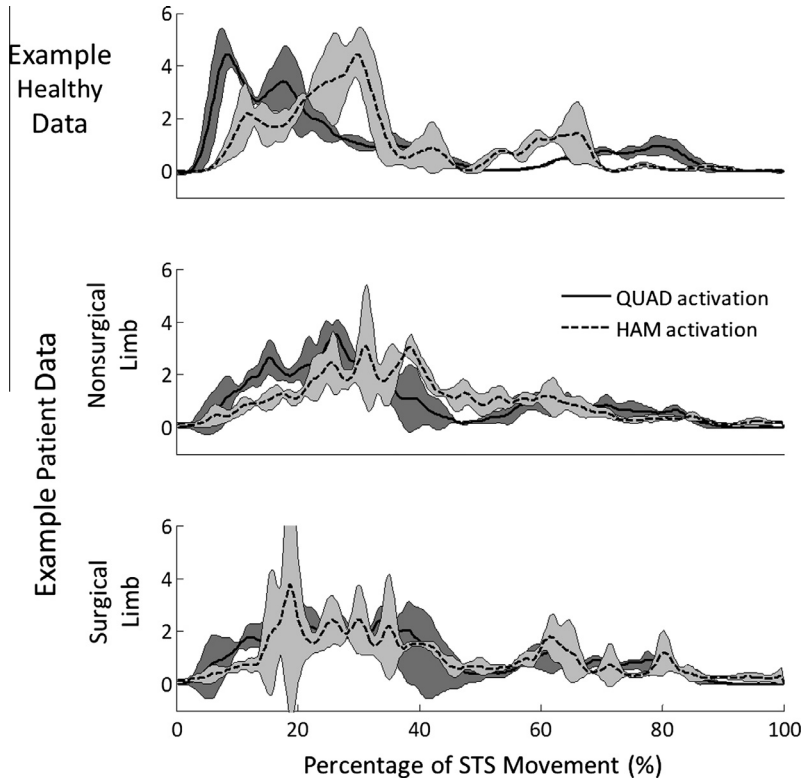

Fig. 2. Ensemble averages of normalized muscle activation (one standard deviation error bounds) from one limb of a representative healthy participant and the postoperative nonsurgical limb and postoperative surgical limbs from a representative patient. The concentric phase occurred from approximately 0–45% and the eccentric phase from 45–100%.

where nEMG is the normalized linear envelope of the EMG signal. Because the linear envelope data were normalized to the mean linear envelope within each repetition, iEMG represents a distribution of recruitment across the phase, and may not be correlated with level of recruitment across the phase.

Coactivation indices (Peterson and Martin, 2010; Winter, 2009) based on the integrated EMG signals were calculated for the concentric and eccentric phases.

$$CI = 2 * \left(\frac{\int_{t_0=0\%}^{t_f=100\%} \min(\mathrm{nEMG_{QUAD}}, \mathrm{nEMG_{HAM}})dt}{\mathrm{iEMG_{QUAD}} + \mathrm{iEMG_{HAM}}} \right) * 100. \quad (3)$$

The patient population had one surgical and one nonsurgical limb. Because this distinction has no meaning in the healthy group, the assignment as nonsurgical and surgical was randomized across all participants in the healthy group. Differences between limbs in the healthy group were examined using two-tailed paired *t*-tests. Dependent variables calculated from force platform and EMG data (iGRF, iEMG, CI) were averaged across each limb in the healthy group for hypothesis testing.

2.5. Data analysis

The primary aim, to compare coactivation *between* limbs in the patient group before and after TKA, was tested using two-tailed paired *t*-tests for each phase. Preoperative and postoperative comparisons were made *within* limb using two-tailed paired *t*-tests for each phase. The secondary aim, to compare coactivation between healthy and patient groups before and after TKA, was tested using a two-tailed independent *t*-test with unequal variance for each phase (concentric, eccentric) and limb (surgical, nonsurgical). Confirmatory measures (iGRF, iEMG, strength, and FTSTS time) were analyzed using identical methods. All statistical calculations were performed with JMP Pro 9 (Cary, NC); alpha was set at 0.05.

3. Results

Age, mass, BMI, and sex distribution were not different between the groups. There was no difference between limbs in the healthy group for any GRF and EMG variable ($P > 0.05$), which allowed for averaging of data across left and right limbs.

Eccentric phase coactivation index (CI) in nonsurgical limb was 32.0% higher preoperatively and 27.0% higher postoperatively than the healthy group (Fig. 3). Eccentric phase CI in the surgical limb was 40.2% higher than the healthy group postoperatively. There were no differences in coactivation between limbs in the TKA group in any phase preoperatively or postoperatively. No differences were found in preoperative integrated ground reaction force (iGRF) when compared to healthy group; however, postoperative iGRF was higher in nonsurgical limb and lower in the surgical limb than healthy group within each phase. There were no preoperative to postoperative differences in CI within either limb.

Compared to the healthy group, postoperative concentric quadriceps iEMG in the surgical limb was 20.7% lower while preoperative quadriceps iEMG in the nonsurgical limb and postoperative quadriceps iEMG in the surgical limb both trended lower (Fig. 4). Concentric quadriceps iEMG in the nonsurgical limb was 15.3% higher preoperatively than the surgical limb. Postoperative eccentric quadriceps iEMG was 52.8% and 74.4% higher than the healthy group in the nonsurgical and surgical limbs, respectively. Preoperative to postoperative comparison in the eccentric phase demonstrated a 31.3% increase in quadriceps iEMG for the nonsurgical limb and an increasing trend for the surgical limb. There were no differences in hamstrings iEMG for any group, time point, or limb comparison.

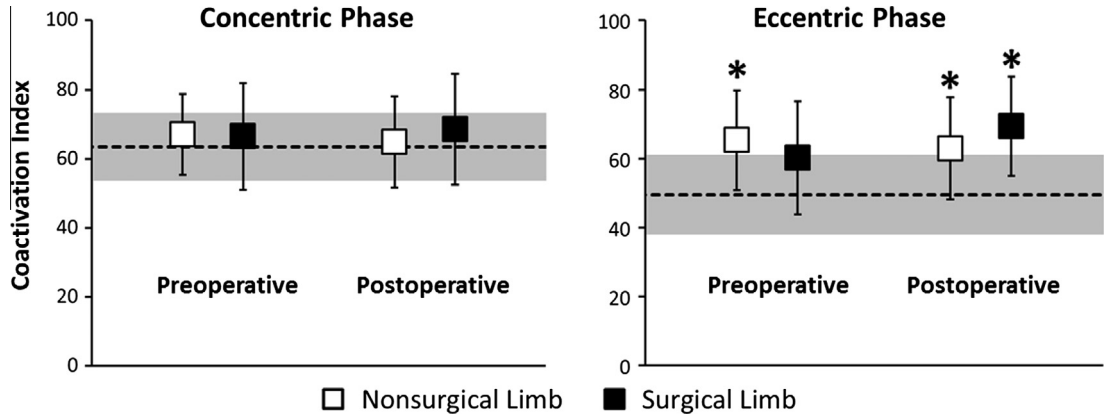

Fig. 3. Mean(SD) coactivation index for the concentric and eccentric phases of the FTSTS task. ∗ indicates $P < 0.05$ when compared to the healthy group ---.

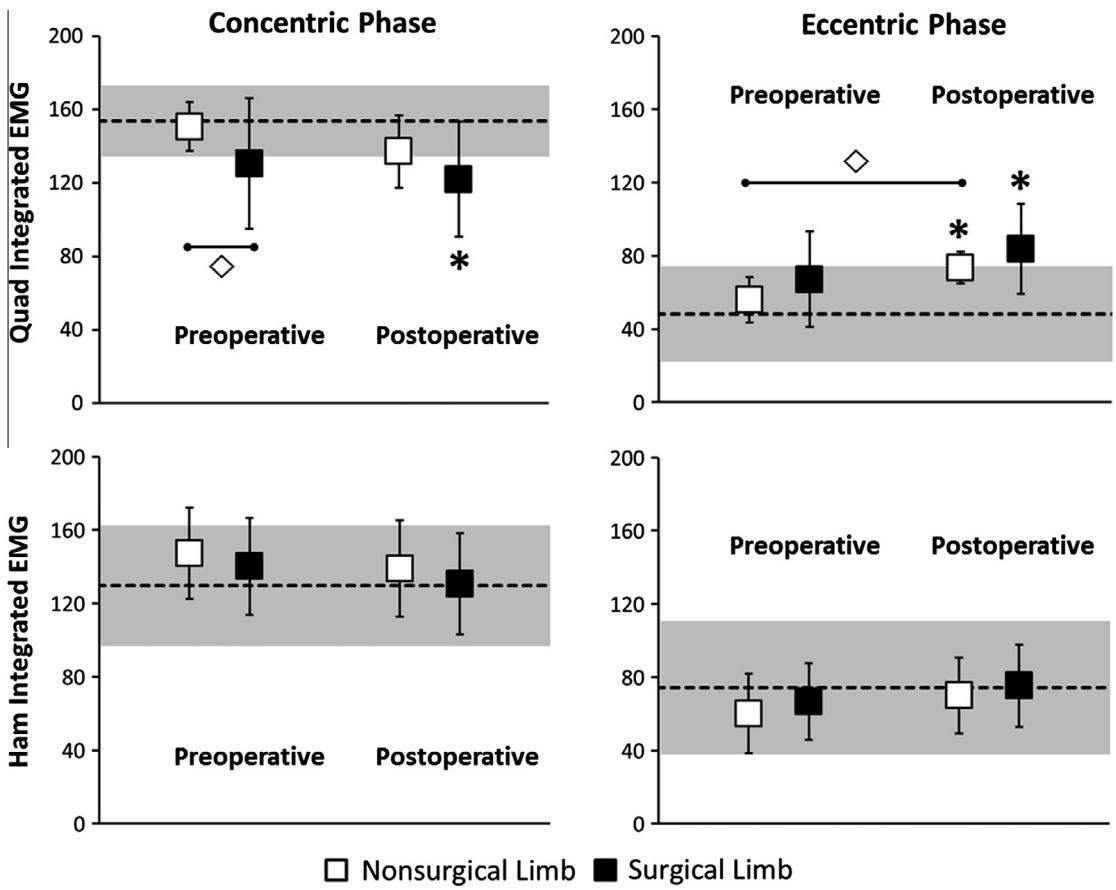

Fig. 4. Mean(SD) of integrated EMG for concentric and eccentric phases of the FTSTS task. ∗ indicates $P < 0.05$ when compared to the healthy group --- and ◇ indicates $P < 0.05$ when compared between limbs or surgical condition.

Limb loading, measured by iGRF, during the concentric and eccentric phases demonstrated similar patterns in concentric and eccentric phases (Fig. 5). Compared to the healthy group, the nonsurgical limb experienced higher iGRF while the surgical limb experienced lower iGRF postoperatively. Surgical limb iGRF was lower than the nonsurgical limb iGRF preoperatively and postoperatively.

Functional outcomes such as FTSTS time and isometric strength testing also demonstrated differences between groups. In the patient group, postoperative FTSTS time did not change. However, the FTSTS time for the healthy group (9.54 (SD 1.88) s) was significantly less than the preoperative FTSTS time (13.06 (SD 2.88) s) and postoperative FTSTS time (15.36 (SD 5.31 s)). Both of these differences are greater than minimum clinically important differences of 2.3 s and 2.5 s reported by Meretta et al. (2006) and Goldberg et al. (2012), respectively. Compared to the healthy group, preoperative and postoperative hamstrings strength was lower for both the nonsurgical and surgical limbs. Preoperative and postoperative quadriceps strength was lower than healthy participants for only the surgical limb (Table 2).

4. Discussion

This investigation assessed a sit-to-stand task and a lowering task in patients undergoing TKA by evaluating ground reaction

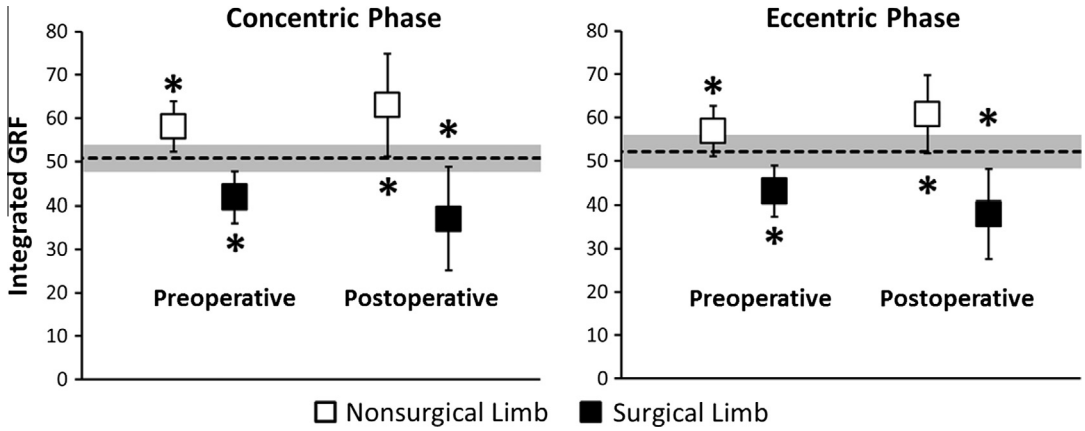

Fig. 5. Mean(SD) of integrated ground reaction force (GRF) for the concentric and eccentric phases of the FTSTS task. ∗ indicates $P < 0.05$ when compared to the healthy group - - -.

forces, time to complete a FTSTS task, and quadriceps and hamstring muscle activation. To our knowledge, this is the first biomechanical investigation that used a fastest possible performance of the FTSTS task both before and early after TKA to evaluate quadriceps and hamstring activation patterns in both the concentric and eccentric phases of the movement. Compared to healthy participants, patients preferentially unloaded their surgical limb, were slower to complete the task, and demonstrated higher quadriceps/hamstrings coactivation with a greater proportion of quadriceps activation when lowering to a seated position. These results highlight areas of focus in postoperative rehabilitation, and may provide additional outcomes to consider when evaluating treatment effectiveness.

As hypothesized, patients who performed a FTSTS task as quickly as possible did so with asymmetric GRFs. Limb loading, measured with vertical GRF, indicated that patients shift the loading demand from the surgical limb to the nonsurgical limb during both concentric and eccentric phases of the FTSTS task. This phenomenon existed both preoperatively and one month postoperatively when compared to healthy adults, and became more pronounced at one month after surgery. In addition, patients took longer to complete the FTSTS task both before and after surgery compared with healthy adult participants, and tended to be slower after surgery than before. These findings are consistent with previous investigations (Christiansen et al., 2011; Christiansen and Stevens-Lapsley, 2010), and highlight the difficulties patients have with functional tasks before and after TKA.

Higher amounts of muscle coactivation were present in the patient group compared with the healthy group during the eccentric phase of the FTSTS task, and indicated an altered strategy for the task. Using these methods, the coactivation index (CI) quantifies how closely matched the distribution of muscle activity is between the agonist and antagonist muscles during each phase of the FTSTS task. Lower CI values in the healthy participant indicates more phasic muscle activity in which quadriceps activity was followed by hamstrings activity during the concentric phase; the pattern was reversed during the eccentric phase (Fig. 2). In contrast, participants from the TKA group adopted a sustained quadriceps recruitment strategy throughout the eccentric phase, which resulted in longer periods of simultaneous activation and higher CI values (Fig. 2). This sustained activity, which is represented by relatively higher quadriceps integrated EMG in the eccentric phase and relative lower quadriceps integrated EMG in the concentric phase (Fig. 4), may be the result of impaired ability to coordinate muscle firing when rapidly lowering to the chair like the healthy group.

Increased distribution of quadriceps activity during the eccentric phase of the task may also indicate that the TKA group employed a protective activation strategy. Despite the fact that eccentric activations can produce higher forces than concentric activation, this increased activity distribution was clearly present in the surgical limb. Anxiety and fear-avoidance beliefs have been associated with self-reported measures of function in patients with knee osteoarthritis (Scopaz et al., 2009) and correspond with clinical observations of cautious movements before and after surgery. Therefore, patients may be overly protective when performing eccentric activities such as lowering to a chair or descending stairs for fear of inducing knee pain, injury, or falling. Our data cannot indicate whether this was a voluntary or involuntary strategy; however, it is logical to increase activation in the muscle that contributes most to regulating the eccentric movement. Conversely, the nonsurgical limb demonstrated muscle coordination of quadriceps activity that more closely mimicked the healthy group. A more normal coordination of muscle recruitment in the nonsurgical limb is particularly important because this limb transmitted most of the ground reaction force during the FTSTS task, which is common in patients with unilateral OA and after TKA.

Both groups demonstrated higher integrated EMG during the concentric phase than the eccentric phase, which corresponds well with execution of the FTSTS task. Participants were directed to complete the FTSTS task as quickly as possible. A large and focused concentric burst of quadriceps muscle activity to achieve a rapid rise from the chair was normal and desired strategy for the task. It appears that the healthy group reduced recruitment of their quadriceps and hamstrings during the lowering phase and allowed gravity to accelerate them to the seated position (Fig. 4). The higher postsurgical quadriceps integrated iEMG in the patients indicates that patients with TKA used a lowering strategy that did not involve a drop to the chair and may contribute to the slower FTSTS times between the groups. One month after surgery, the phasic patterns of quadriceps recruitment in the surgical limb of TKA groups differed even more from the healthy group. At the same time, quadriceps strength was 50% less and hamstring strength was 10% less than preoperative values one month after surgery. This pronounced quadriceps strength loss may be a contributing factor to the differences in EMG recruitment during the FTSTS task for the patient group. The overall movement is less dynamic and unfocused, and patients demonstrate altered muscle recruitment in the surgical limb after surgery compared to before surgery or compared to healthy adults. Patients tended to recruit both quadriceps and hamstrings concurrently during these acute postoperative tests, and distributed recruitment more evenly across phases for both muscle groups (Fig. 2). The higher postoperative coactivation values in the phase are evidence of such a phenomenon.

Table 2
Mean (SD) values for each dependent variable and results of independent *t*-tests comparing the preoperative and postoperative limbs (nonsurgical and surgical) to healthy group (*P*-values, mean difference, 95% confidence interval).

Dependent variable	Healthy	Preoperative				Postoperative			
		Nonsurgical limb		Surgical limb		Nonsurgical limb		Surgical limb	
Coactivation index									
Concentric phase	63.5(11.1)	67.0(11.7)	*P* = 0.47 3.5(−6.6, 13.6)	66.5(15.5)	*P* = 0.61 3.0(−9.3, 15.3)	64.9(13.2)	*P* = 0.79 1.4(−9.5, 12.3)	68.5(16.1)	*P* = 0.42 5.0(−7.7, 17.6)
Eccentric phase	49.5(11.2)	65.3(14.5)	***P* = 0.01 15.7(4.0, 27.4)**	60.2(16.4)	*P* = 0.10 10.8(−2.1, 23.6)	62.9(14.8)	***P* = 0.03 13.4(1.5, 25.3)**	69.4(16.3)	***P* = 0.005 19.9(7.1, 32.7)**
iGRF									
Concentric phase	50.8(2.8)	58.1(5.8)	***P* = 0.004 7.2(2.9, 11.6)**	41.9(5.9)	***P* < 0.001 −8.9(−13.3, −4.6)**	63.0(11.8)	***P* = 0.02 12.1(3.0, 21.3)**	37.0(11.8)	***P* = 0.008 −13.8(−23.0, −4.7)**
Eccentric phase	52.2(3.8)	56.9(5.8)	***P* = 0.04 4.7(0.1, 9.2)**	43.1(5.8)	***P* < 0.001 −9.1(−13.7, −4.7)**	60.8(9.0)	***P* = 0.02 8.6(1.5, 15.7)**	39.2(9.0)	***P* = 0.002 −13.1(−20.1, −6.0)**
Quadriceps iEMG									
Concentric phase	153.9(20.8)	150.7(13.7)	*P* = 0.66 −3.2(−18.2, 11.8)	130.6(35.5)	*P* = 0.09 −23.3(−50.4, 3.8)	137.1(19.8)	*P* = 0.06 −16.9(−34.6, 0.9)	122.0(31.4)	***P* = 0.01 −32.0(−56.5, −7.4)**
Eccentric Phase	48.1(16.6)	56.0(12.3)	*P* = 0.21 7.9(−4.7, 20.4)	67.2(26.1)	***P* = 0.06 19.1(−1.1, 39.3)**	73.5(8.7)	***P* < 0.001 25.4(14.2, 36.7)**	83.9(24.5)	***P* = 0.001 35.7(16.5, 55.0)**
Hamstrings iEMG									
Concentric phase	129.6(31.8)	147.3(24.8)	*P* = 0.15 17.8(−6.8, 42.3)	140.1(26.4)	*P* = 0.40 10.5(−14.7, 35.8)	138.9(26.2)	*P* = 0.45 9.3(−15.9, 34.5)	130.6(27.6)	*P* = 0.94 1.0(−24.9, 26.9)
Eccentric phase	74.3(28.2)	60.2(21.6)	*P* = 0.19 −14.2(−35.8, 7.4)	66.8(20.8)	*P* = 0.47 −7.6(−28.7, 13.7)	70.0(20.7)	*P* = 0.68 −4.3(−25.5, 16.9)	75.4(22.4)	*P* = 0.92 1.0(−20.9, 23.0)
Isometric strength									
Quadriceps	2.01(0.37)	1.63(0.97)	*P* = 0.27 −0.38(−1.09, 0.34)	1.18(0.77)	***P* = 0.009 −0.83(−1.41, −0.25)**	1.66(1.08)	*P* = 0.35 −0.35(−1.14, 0.44)	0.60(0.30)	***P* < 0.001 −1.41(−1.70, −1.12)**
Hamstrings	1.21(0.28)	0.85(0.29)	***P* = 0.007 −0.36(−0.62, −0.11)**	0.67(0.23)	***P* < 0.001 −0.55(−0.77, −0.33)**	0.98(0.30)	*P* = 0.07 −0.24(−0.50, 0.02)	0.60(0.29)	***P* < 0.001 −0.62(−0.87, −0.37)**

The higher levels of quadriceps/hamstrings coactivation early after surgery may accommodate for muscle weakness and inhibition to protect the joint as well as protect the patient from falling while lowering to a chair. However, coactivation is a mechanically inefficient strategy and can hinder functional performance (Damiano, 1993), and the adoption of a chronic coactivation strategy beyond what is necessary to protect the joint early after surgery may be detrimental to muscle strength recovery (Rossi et al., 2006; Silva et al., 2003; Walsh et al., 1998). In particular, the lack of quadriceps isolation during the task may lead to prolonged strength deficits, and may help explain why patients after TKA have difficulty with eccentric activities like sitting down and descending stairs. In addition, the inability to isolate the quadriceps properly, combined with weakness may delay functional recovery up to six months or more following surgery (Bade et al., 2010). Therefore, the early postoperative period may be the opportune time to begin retraining these tasks. Currently, TKA rehabilitation primarily focuses on muscle strengthening and functional training (Akodu et al., 2011; Minns Lowe et al., 2007); however, retraining muscle activation and movement strategies to promote isolated quadriceps recruitment may be a valuable addition to current rehabilitation practice.

Analysis of muscle activity recorded early after TKA during functional tasks has not been previously reported. The results demonstrate the presence of quadriceps/hamstrings coactivation as early as one month after TKA, which is consistent with muscle activity data collected several months after TKA. While examining muscle activity in patients with OA and TKA, previous investigations of gait and stair climbing in adults with knee OA also reported elevated muscle coactivation when compared with healthy older adults during gait (Zeni et al., 2010) and stair climbing. (Hortobagyi et al., 2005). The presence of coactivation with several activities of daily living supports the notion that patients with quadriceps weakness due to knee OA and TKA have difficulty isolating the quadriceps muscle. This may further impair a patient's ability to regain quadriceps strength throughout their recovery from TKA (Rossi et al., 2006; Silva et al., 2003; Walsh et al., 1998). The current work adds depth of understanding related to how the nonsurgical limb musculature is working as a major controller of performance during the task, supporting GRF data obtained in previous studies (Christiansen et al., 2011; Christiansen and Stevens-Lapsley, 2010). It is important to note that the nonsurgical limb does not typically represent a pristine and uncompromised limb for comparison to the surgical limb. Bilateral osteoarthritic disease and muscle weakness are often present in most patients, even in the absence of symptoms, and may affect the strategy chosen to complete this task. The FTSTS task is a bilateral, closed-chain task with some interdependence and influence from the physical impairments present in the surgical knee that could limit the performance of the nonsurgical limb. The ability of the nonsurgical limb to compensate for deficits in the surgical limb is likely maximized in the FTSTS task one month after TKA, which is corresponds to profound quadriceps weakness in the surgical limb.

Several limitations to this investigation should be considered. First, only vertical ground reaction forces were collected; therefore, acceleration only in the vertical direction could be analyzed. Although the FTSTS task is primarily a vertical movement, the horizontal forces and accelerations may be important because the participants were instructed to move quickly during the task. Second, kinematic data were not collected; therefore, joint kinematics were unavailable to guide interpretation of the muscle activity. Finally, it is important to consider that EMG normalization by the mean represents how the participant distributed recruited each muscle across each phase of the FTSTS task, and does not provide information on absolute activation magnitude. The traditional

means of processing involves normalization of the dynamic EMG signal to the maximum signal obtained during a maximum isometric effort. Because the presence of considerable muscle inhibition is common in advanced OA and immediately following TKA, we chose to normalize to the mean EMG signal. Any conclusions made related to the magnitude of intensity of the raw EMG are not possible with this method normalization. However, normalizing by the mean EMG reduces variability (Burden, 2010; Yang and Winter, 1984), and still maintains an appreciation of how patients distribute muscular effort between the quadriceps and hamstrings during performance tests.

5. Conclusion

This investigation demonstrated differing movement strategies during a FTSTS task in patients both before and early after TKA compared to healthy adults. In particular, patients unloaded their surgical limb, were slower to complete the task, and demonstrated higher quadriceps/hamstrings coactivation with increased distribution of the quadriceps activation when lowering to seated position. Although this may represent a protective strategy, it could impede rehabilitation and muscle recovery if it persists. Therefore, retraining movement and activation patterns may be an important component of an effective postoperative rehabilitation program. The current work highlights the distinct muscle recruitment impairments present in patients before and after TKA that may be limiting their physical performance in the task of rising from and lowering to a chair. Future work should investigate why the observed muscle recruitment patterns are adopted, whether similar patterns are present in other activities of daily living, and assess the ability of rehabilitation interventions to address neuromuscular impairments and improve overall physical performance after TKA.

Acknowledgements

This investigation was supported by funding from the National Institutes of Health (K23AG029978 and UL1 RR025780). We also thank Jackie Balter for contribution to this work. We especially thank the participants for their contributions to this investigation.

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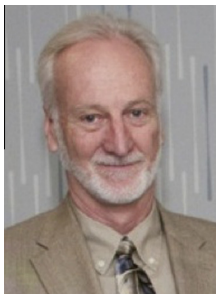

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